

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) Semester III Examination, 2025-26 (NEP)

CHEMISTRY**Paper Code: CHEMJ31 : Inorganic Chemistry-I**

Ref. No.: CECNF/N/2025-26/056 (A)

Time: 2½ Hours

Maximum Marks: 70

Instructions: Answer five questions including Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer all parts:

- (a) Compare the products of reaction when normal-, per- and super oxides of sodium are reacted separately with water. [1½]
- (b) Why do beryllium salts rarely have more than four molecules of water of crystallization associated with the metal ion? [1]
- (c) Write the correct formulation of GaCl₂ indicating the oxidation state of Ga. [1½]
- (d) Write the name and formula of the most widely used nitrogenous fertilizer along with one chemical reaction for its synthesis. [2]
- (e) “The melting and boiling points for NH₃ are out of line as compared to those of its heavier congeners.” Explain, why. [2]
- (f) Write the chemical composition for Portland cement. Why is slow setting of the same desired? How is the same achieved? [2]
- (g) Comment upon the stability of singlet and triplet O₂ along with mentioning of their energy values in kJ/mole. What happens when the singlet O₂ is allowed to react with 1,3-butadiene? [2]
- (h) Write the names of OF₂ and Cl₂O and draw their structures mentioning the hybridization state of the central atom in each case. Why is the Cl-O-Cl bond angle greater than that of F-O-F? [2]

Question 2.

- (a) What do you mean by boranes? Illustrate 2e-2c and 2e-3c kinds of bonds in one lower and one higher borane. Write general molecular formulae for closo-, nido-, arachno- and hypho-boranes. [4]
- (b) Why are Group 1 elements (i) univalent (ii) largely ionic (iii) poor complexing agents? [2]
- (c) Compare multiple bonding in C₂H₄ and P₄O₁₀ giving suitable sketches. [2]
- (d) Why is CO₂ a gas while SiO₂ a solid at room temperature? [2]
- (e) Describe the structure of SO₃ in its solid state. [2]
- (f) Write a balanced chemical reaction showing preparation of H₂O₂. Describe peculiar point of its structure also. [2]

Question 3.

- (a) Comment upon the acidity of H_3BO_3 in aqueous medium. How can its acidity be modulated by (i) taking its higher concentration (ii) adding a compound having cis-diol? [6]
- (b) Draw the structures of C_3O_2 and C_{12}O_9 . Give their preparations also. [4]
- (c) "The dipole moment of NH_3 is much greater than that of NF_3 ." Explain, why. [2]
- (d) How is an electride type complex prepared from Cryptand-222. Comment in very brief upon its magnetic behavior also. [2]

Question 4.

- (a) Give a detailed account of the classification of silicates along with neat sketches of each. [6]
- (b) Discuss in brief the following properties of hydrides of Group 15: (i) Ease of preparation (ii) Stability (iii) Reducing power (iv) Ability to act as electron pair donors. [6]
- (c) " BH_3 undergoes dimerization while BF_3 does not." Explain, why. [2]

Question 5.

- (a) Compare the structure of diamond and graphite. Comment upon their relative thermodynamic stability and conducting behavior also. Describe the preparation of fullerenes from graphite in very brief. [6]
- (b) Discuss the structure of various types of oxides of Nitrogen. Draw the structures of HNO_2 and HNO_3 and mention names along with formulae for their anhydrides. Give a brief account of preparation of HNO_3 through Ostwald's process. [6]
- (c) Write the chemical composition of high alumina cement. Discuss its superiority over Portland cement in very brief. [2]

Question 6.

- (a) What do you mean by Zeolite? Write its general formula. How the same can act as molecular sieve? Support your answer giving a suitable example. [4]
- (b) Define glass and discuss its composition. Give a brief account of different kinds of additions for having glass of desired property. [4]
- (c) Describe the structures for various kinds of oxoacids of sulfur along with mentioning of oxidation state of sulfur atoms in each case. [6]

Question 7.

- (a) Define fertilizer. What are their different types? [4]
- (b) "HF is the weakest acid among all the hydra acids of halogen family in their aqueous state." Explain, why. [2]
- (c) Describe one method by which hydrogen peroxide is prepared. Give the structure of H_2O_2 in the gas phase and mention its peculiar feature. [3]
- (d) Give the names of four different types of oxoacids of chlorine along with their molecular and structural formulae. Comment upon their relative strength giving reasons. [4]

(e) Why is CCl_4 unaffected by water while SiCl_4 is rapidly hydrolysed? [1]
